Enveloppe Live Reaper

Envelop for Live (E4L) in REAPER: quadraphonic spatialisation and live looping

Installation and user guide

> This guide explains, step by step, how to set up a REAPER project where **every audio track moves around four speakers** with its own movement automation, a **master that decodes to quadraphony** (plus a subwoofer), and an **Ableton-style live looper** to build a piece from a synth, all controlled from one window.
> Project repository: **github.com/KinoLab07/enveloppe_live_reaper**

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1. What is included

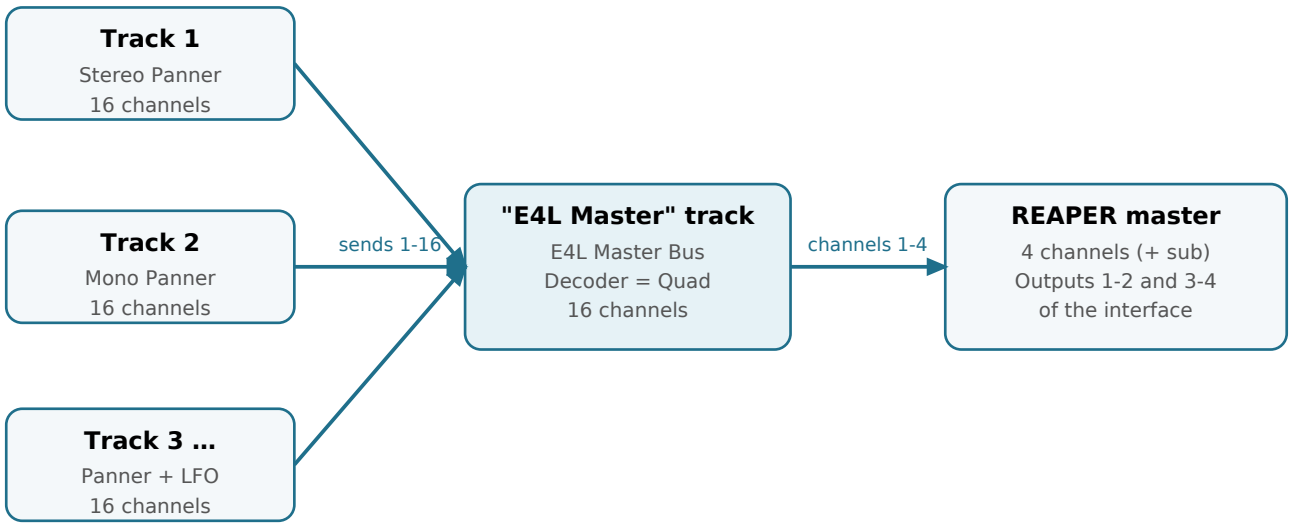
Envelop for Live (E4L) is a set of Ableton Live devices for mixing in surround sound (ambisonics). Those devices only run inside Ableton, so they were **rewritten as native REAPER plug-ins** (JSFX). They run on macOS and Windows, need nothing else installed, and every control can be automated.

Plug-in / script	What it is for
E4L Master Bus	On the spatial master track. Turns the spatial mix into the speaker signals: **Quad**, binaural (headphones), stereo, 5.1, 7.1, 7.1.4 and more.
E4L Mono Panner	Moves a mono source in space, with movement LFOs.
E4L Stereo Panner	Moves a stereo source (left and right apart), with movement and width LFOs.
E4L Quad Panner	Moves audio that is already 4-channel, with LFOs.
E4L Source Panner	Fixed position in degrees or X/Y/Z (no LFO).
E4L Multi-Panner	Up to 16 channels of one track.
E4L Looper	Bar-synced live looper (record, play, overdub, clear).
E4L Sub Out	On REAPER's master: sends everything to the subwoofer output.
E4L - New live project	Script: new project with the whole live set-up and the panel.
E4L - Live Panel	Script: one window to play the loops and move them.
E4L - Update from GitHub	Script: installs the latest version.
E4L - Setup session	Script: spatialises selected tracks (no looper).

- Automated tests check that the sound comes out where it should, that the LFOs turn at the right speed, that tempo sync works and that the looper records exactly on the bar.
- License: **GNU GPL v2 or later**, the same as E4L, since this is a derivative work. You can use, modify and share it as long as the license is kept and Envelop is credited.

2. How it works (the idea in one drawing)

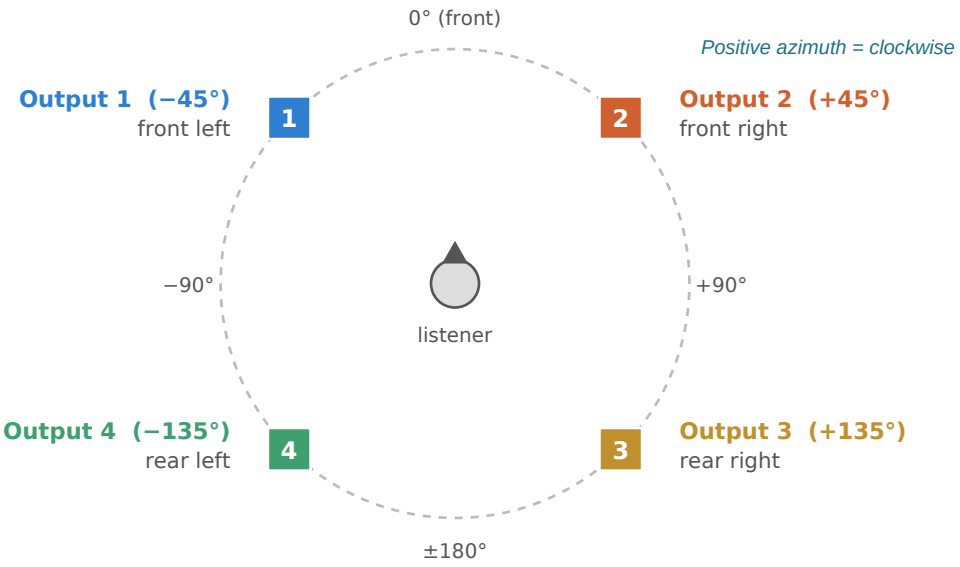
Every audio track carries an **E4L panner**. The panner does not send the sound straight to the speakers but to an intermediate **16-channel** format (ambisonics) that describes *where* the sound comes from. All tracks are sent to the **"E4L Master"** track, where the **E4L Master Bus** turns it into the 4 speaker signals.



Why 16 channels?
Thanks to this intermediate format the same project can be heard in quadraphony, on headphones or on another system by changing **one option** of the Master Bus. The tracks' automation does not change.

Speaker layout

Place the four speakers in a square, at the same distance from the listener, and connect them to the outputs of the audio interface like this:



Mind outputs 3 and 4
E4L's order goes **round the circle, clockwise**: 1 front left, 2 front right, **3 rear right**, **4 rear left**. Many systems use another order (1 FL, 2 FR, 3 RL, 4 RR): if that is your case, wire the speakers following this drawing.

Angles are measured as in E4L: **0°** is the front, **+90°** the right, **−90°** the left and **±180°** the back.

3. Installing in REAPER

Download

1. Open **github.com/KinoLab07/enveloppe_live_reaper** and press the green **Code › Download ZIP** button.
2. Unzip it. You will see the folders **Effects**, **Data** and **Scripts**.

Open REAPER's resource folder

It is where REAPER keeps its plug-ins and scripts. Three ways to open it:

- In REAPER press **?** (action list), type **resource path** and double-click **Show REAPER resource path in explorer/finder**.
- **macOS:** in the Finder press **Cmd + Shift + G** and type **~/Library/Application Support/REAPER**.
- **Windows:** press **Win + R**, type **%APPDATA%\REAPER** and press Enter.

The right folder contains, among others, the folders **Effects**, **Data** and **Scripts**, and a **reaper.ini** file.

Copy the files

1. Close REAPER.
2. Copy **Effects**, **Data** and **Scripts** from the zip into the resource folder. When asked about **folders**, choose **merge**; about **files**, choose **replace**. Never replace a whole folder, or you will lose your other plug-ins.
3. Open REAPER, open the FX browser (**FX** button of a track) and press **F5** (Mac laptops: **fn + F5**). Type **E4L**: the **JS: E4L …** plug-ins appear.

Load the scripts (once)

1. Open **Actions › Show action list…** (or press **?**).
2. Bottom right: **New action…** › **Load ReaScript…**
3. In the resource folder choose **Scripts/E4L/E4L - New live project.lua** and press **Open**. Repeat with **E4L - Live Panel.lua** and **E4L - Update from GitHub.lua** (and any other script you want to use).
4. Optional: with a script selected press **Add…** (bottom left) and press a key, e.g. **Shift + E**, to run it with a shortcut.

> **Updating (macOS and Windows)**
> Run **E4L - Update from GitHub** whenever there is a new version. It downloads it from GitHub and replaces the plug-ins, scripts, filters and this guide in the resource folder, without touching your other plug-ins. It tells you when you are already up to date.
> Then it **closes and reopens the Live Panel** by itself and offers to **reopen the project**, which is how the new plug-ins are loaded (loops recorded in memory are lost). The updated guide is in **Data/E4L** inside the resource folder.
> Only if a **new** plug-in does not show up in the FX browser: with that window active press **F5** (**fn + F5** on a Mac laptop).

4. Setting up a project for quadraphony

This section explains the set-up by hand, for normal (non-looping) projects. For live looping, **E4L - New live project** does all of it in one click (section 8).

Step A: the audio interface

1. Open **Settings › Audio › Device** (Windows: **Options › Preferences › Audio › Device**).
2. Choose your interface and make sure at least **outputs 1 to 4** are enabled.

Step B: REAPER's master to 4 channels

1. On the **MASTER** track press **Route** (or right-click the master › **Routing**).
2. At the top, in **Track channels**, choose **4** (or **6** with a sub on output 5).
3. In **Audio Hardware Outputs** keep the existing output: channels **1/2 → outputs 1/2**.
4. Press **Add new hardware output**, choose **outputs 3/4** and set its source to channels **3/4**.

Step C: the "E4L Master" track with a script

1. Select the audio tracks to spatialise (click the name; **Cmd/Ctrl + click** for several).
2. Run **E4L - Setup session** (press **?**, search "E4L", double-click).
3. Type the panner number and press **OK**: **1** Stereo, **2** Mono, 3 Source, **4** Quad, 5 Multi.

The script creates the **"E4L Master"** track with **E4L Master Bus** and gives every selected track **16 channels**, the panner, its master send **off** and a 16-channel send to "E4L Master".

Step D: the Master Bus in quadraphony

1. On "E4L Master" open **E4L Master Bus** (**FX** button).
2. Set **Decoder** to **Quad**.
3. Keep **Order Weighting** on **In-phase (E4L)** and **Ambisonics Order** on **3rd order**.

> **Why "In-phase"?**
> The Quad decoder was measured with the three *Order Weighting* options. **In-phase (E4L)** is the best for four speakers: when a sound turns the level stays almost constant and no speaker gets an inverted signal. With *Basic* or *max-rE* the level drops a lot between speakers.

Check

- "E4L Master" has **16 channels** and its master send **on**.
- Audio tracks have **16 channels**, the **master send off** and a send to "E4L Master" with **Audio: 1-16 → 1-16**.
- When playing, **4 bars** move on the master meter.

5. Adding audio tracks and positioning them

Which panner?

Kind of audio	Plug-in	Notes
Mono (a voice, an instrument)	**E4L Mono Panner**	One moving point.
Stereo	**E4L Stereo Panner**	Two points (L/R) moving together; their distance is the *Azimuth Spread*.
Already 4 channels	**E4L Quad Panner**	Four points moving together; audio on channels 1-4 of the track.

The basic controls

Control	What it does	For quadraphony
Azimuth	Direction in degrees: 0 front, +90 right, −90 left, ±180 back.	−45 = speaker 1, +45 = speaker 2, +135 = speaker 3, −135 = speaker 4.
Radius and **Range**	Distance: Radius × Range. Further = quieter (−1.5 dB per unit).	**Radius = 0.1** with Range 10 for full level.
Elevation	Height in degrees.	Keep it at **0**: quad has no speakers above.
Azimuth Spread	(Stereo/Quad) Width between the channels.	Stereo: 30 = narrow; 45 = L and R on speakers 1 and 2 with Azimuth 0.
Azimuth Spread Mode	(Quad) LR = square; X = rear pair crossed; W = fan.	With LR and spread 90 each channel goes to its speaker.

Too quiet?
With the factory values (Radius 1, Range 10) the sound is **13.5 dB down**: that is how E4L works, it treats it as "far". For normal level set **Radius to 0.1**. You can automate Radius to bring the sound closer and further.

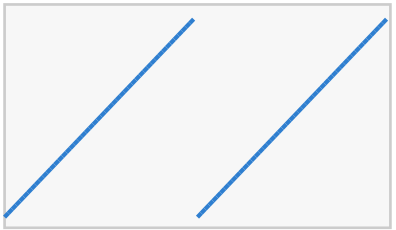
6. Movement automation

There are two ways to move a sound, and they can be combined:

- **Built-in LFOs** (as in E4L): automatic repeating movements (turns, swings, tempo-synced movements). Just switch them on. The Live Panel's **MOVEMENT** editor shows them with plain names (section 8).
- **REAPER envelopes**: you draw the path on the timeline, point by point.

6.1 Built-in LFOs

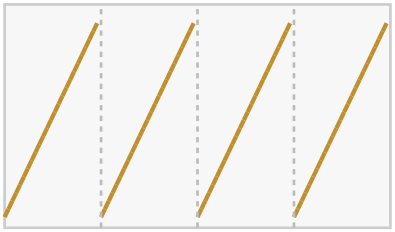
Cycle: full turn



Sin: swing ± Depth



Sync: 1 turn every N bars



Azimuth LFO control	What it does
Azimuth LFO On/Off	Switches the automatic movement on or off.
Mode	**Cycle**: full 360° turns around the 4 speakers. **Sin**: swings around *Azimuth*, ± *Depth* degrees.
Time Mode	**Freq**: speed in Hz (turns per second). **Sync**: synced to the project tempo.
Frequency and **Rate**	(Freq) Speed = Frequency × Rate. A negative Rate turns the other way. Example: 0.1 Hz × 0.5 = one turn every 20 s.
Sync (bars)	(Sync) Bars per turn: 1/4, 1/2, 1, 2, 3, 4, 6, 8, 16, 32, 64.
Direction	(Sync) Clockwise or Counter.
Depth	(Sin) Swing width in degrees.
Azimuth	With the LFO on: start point (Cycle) or centre of the swing (Sin).

Ready-made recipes

Movement	Settings
Slow turn around the room	LFO On · Mode **Cycle** · Time Mode **Freq** · Frequency **0.1** · Rate **1** (10 s per turn). Other way: Rate −1.
One turn every 4 bars	LFO On · Mode **Cycle** · Time Mode **Sync** · Sync **4** · Direction as you like.
Swing between the two front speakers	Azimuth **0** · LFO On · Mode **Sin** · Depth **45** · Frequency 0.25 (4 s there and back).
Swing front to back on the right	Azimuth **90** · Mode **Sin** · Depth **45** (45° to 135°, between speakers 2 and 3).
Side to side through the front	Azimuth **0** · Mode **Sin** · Depth **90**.
A stereo source that "breathes"	Stereo Panner · **Azimuth Spread LFO On** · Frequency 0.2 · Depth 30.

Tempo sync
In **Sync** mode, while REAPER plays, the movement is locked to the project position: the sound is always in the same place at the same bar, on every playback and in the render.

6.2 REAPER envelopes (drawing the path)

1. On the track press the **envelope** button (the curve icon next to **FX**).
2. Find the **E4L** panner and tick **Azimuth**. An automation lane appears.
3. **Click** on the lane to add points, **drag** them to move them. Top = +180°, middle = 0° (front), bottom = −180°.
4. Right-click a point to change the curve shape (linear, smooth, square…).
5. Same for other parameters: **Radius** (closer / further), **Azimuth Spread** (width).

Recording the movement live: set the track's automation mode (usually **Trim/Read**) to **Latch**, play and move **Azimuth** with the mouse; then set it back to **Read**.

> **Crossing the back**
> +180° and −180° are the same point (back). A line from +170 to −170 goes through every value in between, so the sound **goes round through the front**. For full turns use the LFO in **Cycle** mode, or a point at +180 followed by −180 with a **square** shape.

6.3 Combining LFOs and envelopes

- Automate **Azimuth LFO On/Off** so the turn starts at bar 9 and stops at bar 17.
- Automate **Azimuth LFO Frequency** so the turn speeds up little by little.
- Automate **Azimuth** with the LFO in **Sin** so the centre of the swing travels.
- REAPER's **Parameter modulation** (in the plug-in: **Param › Parameter modulation**) can also drive any control, e.g. from the level of another track.

7. Headphones and export

Preview on headphones

Without the four speakers at hand, set the **E4L Master Bus Decoder** to **Binaural (3rd-order)**: outputs 1-2 then carry a 3D headphone mix (front, back and sides). **Set it back to Quad** before exporting or using the speakers. (The live set-up of section 8 already has a separate binaural CUE track.)

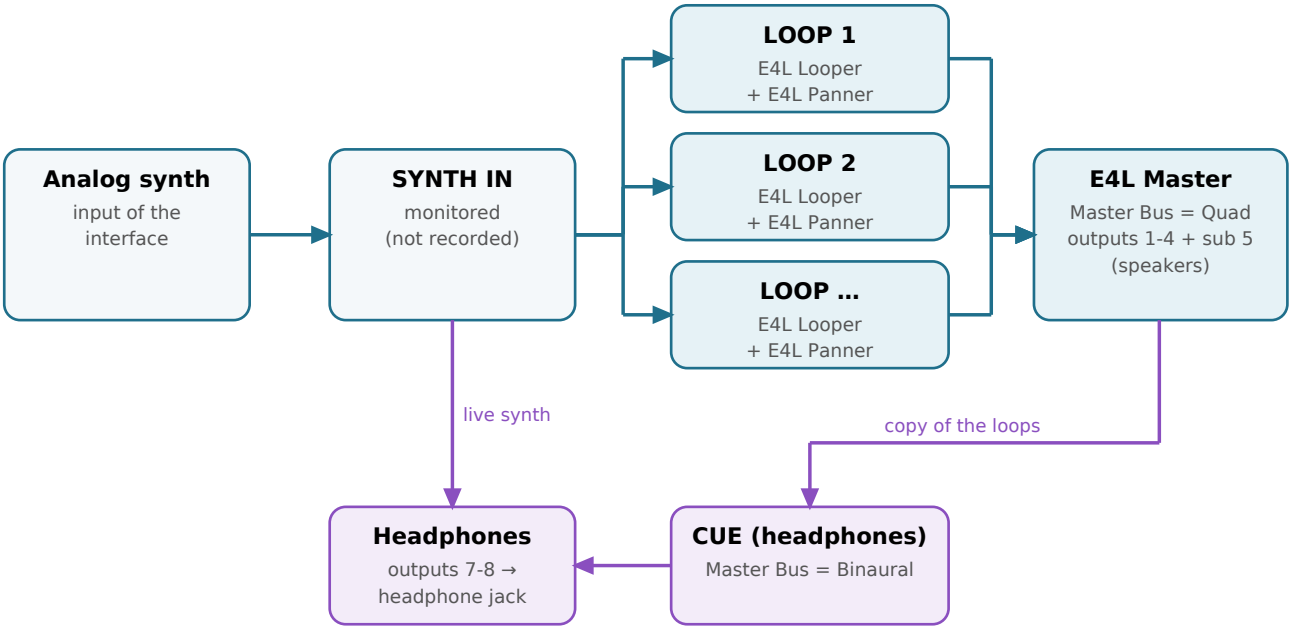
Exporting the quadraphonic file

1. **File › Render…**
2. **Source:** Master mix.
3. **Channels: 4** (**6** to include the sub on channel 5).
4. **WAV** format, then **Render**.

Channel order (E4L): **1** front left, **2** front right, **3** rear right, **4** rear left.

8. Live looping, Ableton style (analog synth)

Compose live: you play the synth and hear it **only on the headphones**. When you like a sound you record it into a **loop of N bars** that starts **exactly on the next bar**, in sync with the "global loop" (the project's tempo and bars). From then on it plays on the speakers, in its own position, with its own movement.



Before you start: headphones and sub (Mbox Pro)

The headphones need a mix **different** from the speakers'. The set-up sends the headphone mix to **outputs 7-8** of the interface, which feed no speaker. On an **Avid Mbox Pro (3rd gen)** each headphone output (A and B) has its own mix, set in the driver's control panel:

1. In REAPER open **Settings › Audio › Device** and open the interface's control panel, or open the **Mbox Pro Control Panel** directly.
2. In the output settings, set **Headphones A** (and B if used) to what comes from the computer on **outputs 7-8**. Exact names depend on the driver version.
3. Check that outputs **1-4** (speakers) and **5** (sub) carry what comes from the computer on those same outputs.
4. The Mbox master volume affects every output; to make it control the headphones only, untick *Master Vol.* for the other outputs in **Output Setup**.

> **The sub (output 5)**
> The set-up puts **E4L Sub Out** on **REAPER's master**: the sum of **everything that reaches the master** also goes to **output 5**, so **any track you add** feeds the sub.
> In the plug-in: **Sub Level** and **Sub Low-pass** (0 = no filter, if the sub has its own crossover; otherwise 80-120 Hz).
> For an existing project run **E4L - Add sub to master** and type the sub output (5).

Creating the live project

Run **E4L - New live project**: a new project, the whole live set-up and the **Live Panel**, in one click (the same as running **E4L - Live Looper template** and then **E4L - Live Panel**). Fill in the dialog and press **OK**:

Field	What to type	Default
Loop tracks to add	How many loop tracks.	8
Synth input (first channel)	Interface input where the synth is connected.	1
Synth input: 1 mono 2 stereo	1 if the synth has one output, 2 if it has two.	1
Headphones output (first channel)	First output for the headphones.	7 (outputs 7-8)
Loop panner: 1 Mono 2 Stereo	Panner of each loop: Mono for a mono synth.	1
Sub output (0 = none)	Subwoofer output; it gets the sum of everything.	5

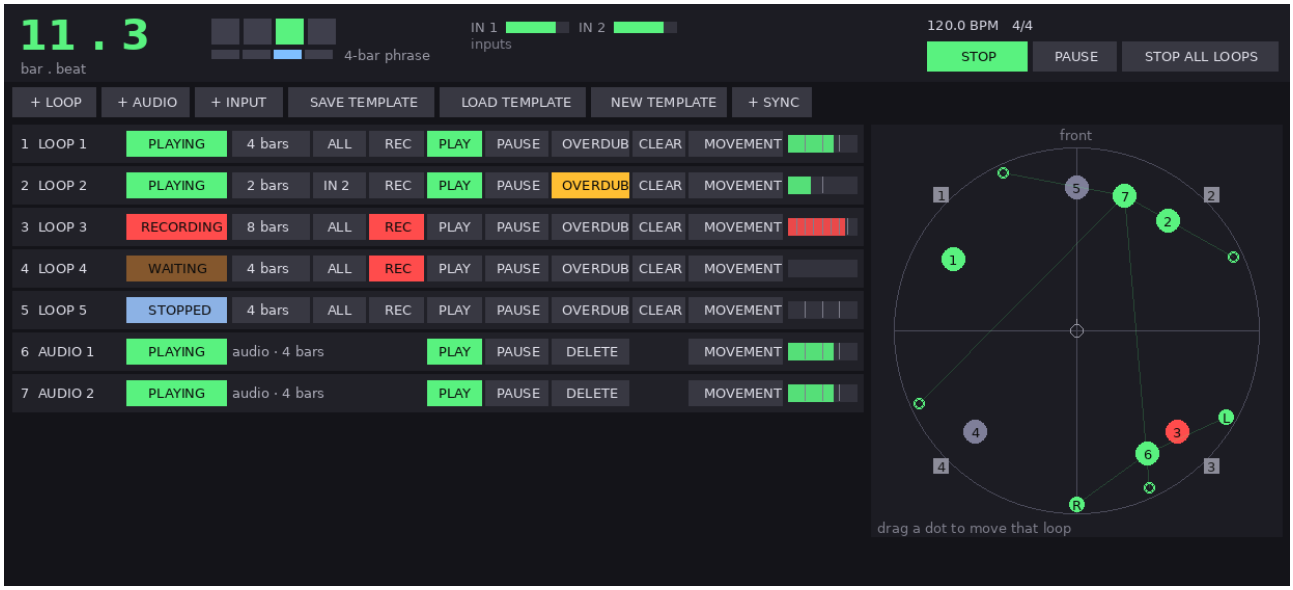
It creates **SYNTH IN**, **LOOP 1…8**, **E4L Master** (quad) and **CUE (headphones)**, sets up the master (outputs 1-2, 3-4, sub on 5) and measures your interface's **recording latency compensation**. Each loop starts in a different place in the room (−45°, 45°, 135°, −135°…).

Playing live, step by step

1. Set the **tempo** and **time signature** (box below).
2. Press **PLAY** in the panel (or the space bar). The running project is the "global loop": every loop follows its bars.
3. Play the synth: you hear it on the headphones. Nothing goes to the speakers yet. The **IN** meters at the top of the panel show that the input reaches REAPER.
4. On a loop row choose the length (click it: 1, 2, 4, 8, 16 or 32 bars) and press **REC**.
5. The state shows **WAITING**; on the next bar it becomes **RECORDING** (red) and records exactly the chosen bars.
6. Then it switches to **PLAYING** (green) by itself: the loop plays on the speakers and on the headphones (CUE).
7. For the next sound use the next loop: each layer on its own track, in its own place.

Setting the tempo and time signature
Transport bar (below the tracks): on the right you see the tempo (**BPM**, e.g. 120) and the time signature (e.g. **4/4**). **Double-click** them to type a new value, or drag the BPM up and down.
Project settings: File › Project settings…: *Project BPM* and *Time signature*.
Tap tempo: the action *Transport: Tap tempo*.
Metronome: the metronome button in the transport bar. To hear it on the headphones only, send it to outputs 7-8 in **Settings › Metronome**.
Set the tempo **before** recording loops: changing it later speeds up or slows down the recorded loops (and changes their pitch).

The Live Panel: everything in one window



- **Top:** the **bar . beat** counter, the beats of the bar (the first one in orange), a **4-bar phrase** indicator, the **IN** input meters, the tempo, PLAY/STOP, PAUSE and **STOP ALL LOOPS**.
- **STOP** (or the space bar) stops and goes back to bar 1: on PLAY every loop and audio file starts again from its beginning, like Ableton. **PAUSE** keeps the position: PLAY continues exactly where it was.
- **Toolbar: + LOOP**, **+ AUDIO**, **+ INPUT**, **SAVE TEMPLATE**, **LOAD TEMPLATE**, **NEW TEMPLATE** (see below), **+ SYNC** (reset pulse for an external sequencer, see below).
- **Syncing an analog sequencer:** the **SYNC OUT** track (created by the template or by **+ SYNC**) sends a short pulse to output 6 of the Mbox on the bar where a loop starts recording, and when REAPER starts from STOP. Patch output 6 to the **RESET** input of the sequencer: the sequence starts on step 1 exactly with the loop. In its window (button **SYNC**): **SEND A RESET** to test the cable, pulse length, level, polarity, a timing offset, reset every N bars, a clock on output 2 and MIDI Start / clock. If the sequencer does not react, try a longer pulse or Polarity = Negative (interface outputs give little voltage).
- **One row per loop:** state (EMPTY, WAITING, RECORDING, PLAYING, STOPPED, PAUSED; it blinks while waiting for the bar), length, input (**ALL** or **IN n**), **REC**, **PLAY**, **PAUSE**, **OVERDUB**, **CLEAR**, **MOVEMENT** and the position inside the loop.
- **The room seen from above:** the 4 numbered speakers and one dot per loop (green playing, red recording), moving live with the animations. **Drag a dot** to move that loop.
- **Keys: Space** = play/stop · **1-9** = record on loop 1-9 · **0** = stop all loops · **Esc** = close the movement editor.
- **Right-click the top area** to **dock** the panel inside REAPER.

Loop button	What it does
REC	Records on the next bar. On a loop that already has audio it records a new one; the old one keeps playing until the new recording starts.
PLAY	Starts the loop from its beginning on the next bar, or stops it on the next bar.
PAUSE	Silences the loop at once. Press it again: it continues at once from the exact point where it was paused, like a tape (it moves against the bars by the time it was paused). PLAY puts it back on the bars, from its beginning.
OVERDUB	While on, what you play is added on top of the loop.
CLEAR	Erases the loop.
ALL / IN n	Which interface input the loop records: **ALL** (every input, the default) or only **IN 1**, **IN 2**... Click to change.
MOVEMENT	Opens the movement editor of the loop.

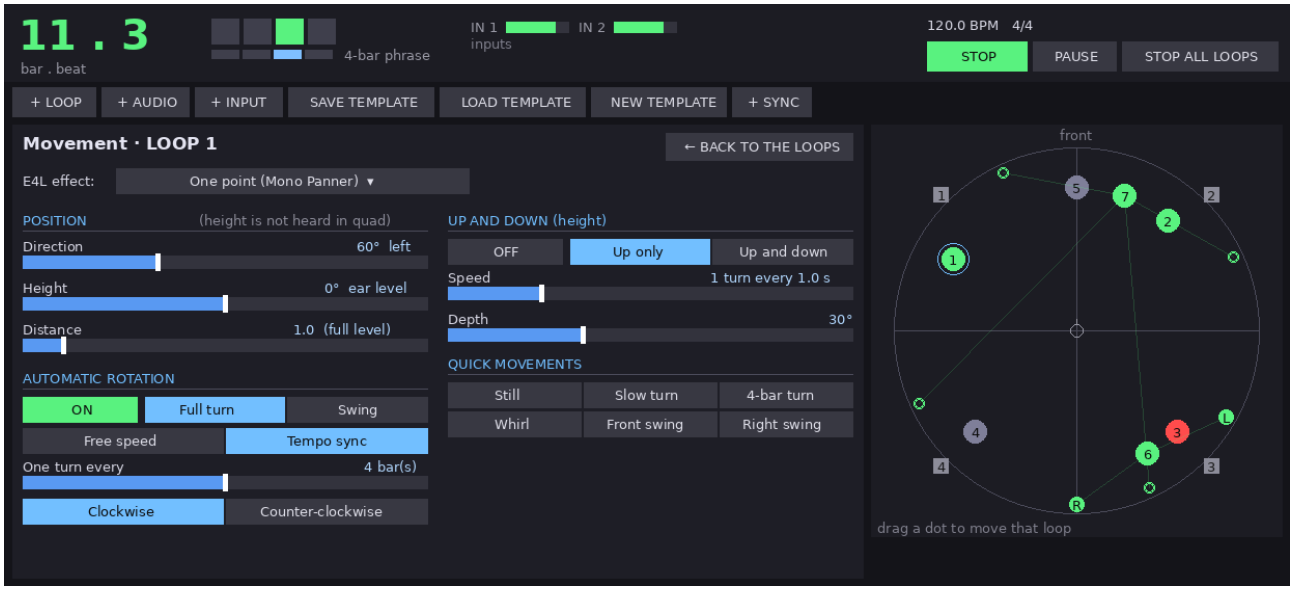
Pre-recorded audio, more inputs and more loops

- **+ AUDIO**: choose a file (wav, aif, mp3...). A new **AUDIO n** track is created, spatialised like a loop. The file starts at bar 1 and repeats; if it lasts almost a whole number of bars it is fitted to the tempo without changing its pitch. It starts **STOPPED**: press **PLAY** and it comes in from its beginning on the next bar (same to take it out). **PAUSE** works as on the loops: silent at once, then it continues from the bar where it was paused. It has its own **MOVEMENT** button. You can also drag files from the Finder onto the panel.
- The effect follows the channels of the file: **mono** = one point; **stereo** = two points, L and R on speakers 1 and 2; **quad** = four points, each channel on its own speaker (channel order FL, FR, RL, RR); **5 or more** channels = Multi-Panner, in a circle. The room view shows one small dot per channel. Turning the direction or the automatic rotation turns the whole image.
- **DELETE** on an audio row: click once (it turns into **SURE?**) and again within 3 seconds. Cmd+Z in REAPER brings it back.
- **+ INPUT**: type the interface channel and mono/stereo. A new **SYNTH IN 2** (then 3...) is created; it also goes to the headphones and has its own **IN** meter. Loops set to **ALL** record it straight away.
- **+ LOOP**: another loop track at the end, ready to record.

Templates: save, load and move to the next piece

- **SAVE TEMPLATE**: give it a name; it is saved in **E4L_Templates** (in REAPER's resource folder) with its tracks, inputs, audio files, positions, movements and tempo.
- **LOAD TEMPLATE**: lists the saved templates and opens the one you choose (REAPER asks first if the current one has unsaved changes).
- **NEW TEMPLATE**: a new live project from scratch.
- For a concert: prepare one template per piece. Live, at the end of a piece press **LOAD TEMPLATE** and choose the next one. Switching causes a short audio gap, and **loops recorded live are not saved**: each template starts with empty loops.

The movement editor



Part of the editor	What it does
E4L effect (menu)	**One point** (Mono Panner), **Two points, left and right** (Stereo Panner) or **Fixed position** (Source Panner, no animation). Switching keeps the position and the animations.
Position	**Direction** (e.g. "45° right"), **Height** and **Distance** (shown as dB of attenuation).
Automatic rotation	ON/OFF · **Full turn** or **Swing** · **Free speed** ("1 turn every 10 s") or **Tempo sync** ("one turn every 4 bars") · **Clockwise / Counter-clockwise** · **Swing depth**.
Up and down	Height animation: **Up only** or **Up and down**, speed and depth (barely audible in quad: no speakers above).
Stereo width	(Two points only) **Left / right spread** and **Breathing**: the image opens and closes by itself.
Quick movements	One click: **Still**, **Slow turn**, **4-bar turn**, **Whirl**, **Front swing**, **Right swing**.

Keyboard and MIDI control

- **One button per loop (best with a controller):** in a track's **E4L Looper** press **Param › Learn**, choose **Record** (or **Play**) and press the controller button. Repeat for each loop.
- **Keyboard:** load **E4L - Loop REC (selected tracks)**, **E4L - Loop PLAY-STOP (selected tracks)** and **E4L - Loops STOP all** and give them keys (Actions › Add).

Good to know
• REAPER must be **playing**: the looper follows the project position. Do not use REAPER's repeat unless its length is a multiple of the loops' length.
• Loops **live in the plug-in's memory** while REAPER is open: **they are not saved with the project**. Render (section 7) to keep a performance.
• Maximum loop length: about 5 minutes at 48 kHz (plenty for 32 bars).
• To hear the synth without the computer's latency, use your interface's **direct monitoring** and turn the SYNTH IN track down.
• If a loop is slightly early or late, adjust **Recording Latency Compensation** in its E4L Looper (more ms if it sounds late).
• Steady, unmodulated tones (drones, bass): a wave whose cycles do not fit the loop exactly would jump in phase at the seam. Right after recording, the looper finds the shift (at most half a cycle) that puts the end in phase with the start and reads the loop that much longer or shorter: a few cents, spread over the whole loop, and a seam without dip or click. Sounds that are not periodic keep their exact speed. Keep the note sounding a moment before REC starts: the crossfade uses that sound.

9. Troubleshooting

Problem	Solution
The E4L plug-ins do not show up	Check that **Effects/E4L** is inside the resource folder and press **F5** (fn + F5) in the FX browser.
Nothing is heard	The track must have **16 channels** and a send to "E4L Master"; "E4L Master" must have its **master send** on. Check that Radius is not too high.
Too quiet	Set **Radius** to 0.1 (section 5).
Only 2 speakers play	REAPER's master needs **4 channels** (6 with the sub) and hardware outputs 1/2 and 3/4 (section 4, step B). The Master Bus must be on **Quad**.
The rear speakers are swapped	Output 3 = rear **right**, output 4 = rear **left**. Swap those two cables.
The sound is also heard unspatialised	The track's **master send** is on: switch it off (Route › Master send).
A second input is heard but not recorded	The loop's input button must show **ALL** or the right **IN n**. Check the **IN** meter of that input at the top of the panel.
The looper says "Press play in REAPER"	Press PLAY: the looper only records and plays while the project runs.
The synth comes out of the speakers	SYNTH IN must have its master send off, and the looper's **Input Thru** must be off.
The synth is not on the headphones	SYNTH IN must be armed (red button) with monitoring on; in the Mbox panel the headphones must take outputs 7-8 (section 8).
The sub is silent	REAPER's master needs **E4L Sub Out**, 6 channels and a mono hardware output to output 5: **E4L - Add sub to master** sets it all up.
The panel did not change after an update	Close the panel and run **E4L - Live Panel** again (the new updater does it by itself).
A script shows an error	Copy the error text or take a screenshot and send it.
Master Bus says "HRIR filters not found"	Binaural only: **Data/E4L/hrtf** is missing in the resource folder.

10. Quick parameter reference

E4L Master Bus

Parameter	Values	Recommended here
Decoder	Binaural, Stereo, Quad, Octo, 5.1, 7.1, 5.1.2, 7.1.4, AmbiX	**Quad**
Order Weighting	In-phase (E4L), max-rE, Basic	**In-phase (E4L)**
Ambisonics Order	3rd order (16 ch), 1st order (4 ch)	**3rd order**
Input / Output Gain	−70 to +6 dB	0 dB
Sub Output	Decoder's own sub (0 = off)	**0**: the sub is on the master (E4L Sub Out)

E4L Mono / Stereo / Quad Panner

Group	Parameters
Position	Azimuth (−180…180), Elevation (−90…90), Radius (0…1), Range (1…40)
Width (Stereo/Quad)	Azimuth Spread, Elevation Spread, Azimuth Spread Mode (Quad)
Azimuth LFO	On/Off, Mode (Cycle/Sin), Time Mode (Freq/Sync), Frequency, Rate, Sync, Direction, Depth
Elevation LFO	On/Off, Polarity (Unipolar/Bipolar), Time Mode, Frequency, Rate, Sync, Direction, Depth
Spread LFOs (Stereo/Quad)	On/Off, Mode (Freq/Sync), Frequency, Sync, Depth

E4L Sub Out (on REAPER's master)

Parameter	Values	Recommended
Sub Output (channel)	1 to 16	**5**
Sum of	Channels 1-4 (quad), 1-2 or 1-6	**1-4**
Sub Level (dB)	−40 to +12	0 dB; adjust by ear
Sub Low-pass (Hz)	0 = off, or 40-250	0 if the sub has a crossover; else 80-120
Sub	On / Muted	On

E4L Looper

Parameter	Values
Length (bars)	1, 2, 4, 8, 16, 32
Record / Play / Overdub / Clear	Buttons, quantised to the bar
Loop Volume (dB)	−60 to +12
Input Thru	Off / On
Recording Latency Compensation (ms)	0 to 500 (set automatically)
Loop Seam Crossfade (ms)	1 to 170 (50)
Seam Level Compensation	Auto (by pitch) / Off / Full
Start / Stop Fade (ms)	1 to 200 (30)

E4L Sync Out

Parameter	Values
Reset when a loop starts recording	On / Off (On)
Reset when REAPER starts from STOP	On / Off (On)
Reset every N bars	Off, 1, 2, 4, 8, 16 bars
Pulse Length (ms) / Level / Polarity	1-100 (10) / 0-1 (1) / Positive, Negative
Timing Offset (ms)	−50 to +50 (negative = earlier)
Clock on output 2	Off, 1/4, 1/8, 1/16, 1/32, 24 PPQN
MIDI	Off, Start on reset, Start + clock 24 PPQN

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